

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (AIML) (Credit System) Degree Examinations

May - June 2023

20AM602 - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING IN HEALTHCARE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

	Marks	BT*	CO*	PO*
1. a) What is AI in Health care? Explain the use of AI in Health care.	10	L*2	1	1
b) With block diagram explain the Framework for AI in disease detection modeling.	10	L2	1	1
2. a) With block diagram explain the diagnosis of Alzheimer's disease.	6	L2	1	1
b) With block diagram explain Heart disease diagnosis.	6	L2	1	1
c) Explain types of clinical information system.	8	L2	1	1
3. a) What is cross validation? Explain different methods of cross validation.	10	L2	2	1
b) With block diagram explain Automated Machine Learning.	10	L2	2	1

Unit – II

4. a) What is image classification? With block diagram explain digital image classification phases.	10	L2	3	1
b) With diagram explain The CAESAR-ALE Framework.	10	L2	3	1
5. a) With block diagram explain the methods of MRI image segmentation.	10	L2	3	1
b) With block diagram explain the supervised classification used in Image Classification.	10	L2	3	1
6. a) Explain different Approaches to knowledge representation and types of knowledge.	10	L2	4	1
b) Explain the techniques of knowledge representation.	10	L2	4	1

Unit – III

7. a) Explain fuzzification and defuzzification.	6	L2	5	1
b) With diagram explain the architecture of fuzzy system.	6	L2	5	1
c) What is genetic algorithm? List its advantages and limitations.	8	L2	5	1
8. a) With block diagram explain the basic structure of genetic Algorithm.	6	L2	5	1
b) With flow chart, explain Working Mechanism of Genetic Algorithms.	10	L2	5	1
c) Explain Two Armed And K – Armed Bandit Problem.	4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
